

## TOPICS FOR TODAY:

### Link Layer

node to node delivery of frames  
↳ access control  
↳ flow control  
↳ error control

- CRC algorithms - refer recorded video
- Address Resolution Protocol (ARP)
- Reverse ARP (RARP)
- Access Control

### Misc

- End-to-end example

↳ how different protocols/technologies are used in the Internet

\* Have you learnt everything about Modern Networking?

# \* Link layer: Address Resolution Protocol (ARP)

maps IP address to MAC addr

Case 1:

ARP req (A → B) → IP addr  
src: A  
dst: ff:ff:ff:ff:ff:ff

ARP reply  
B's MAC

src: B  
dst: A

route → n

Cache into ARP table

Case 2:

A → B

default gateway

ARP req  
i1's MAC  
ARP reply

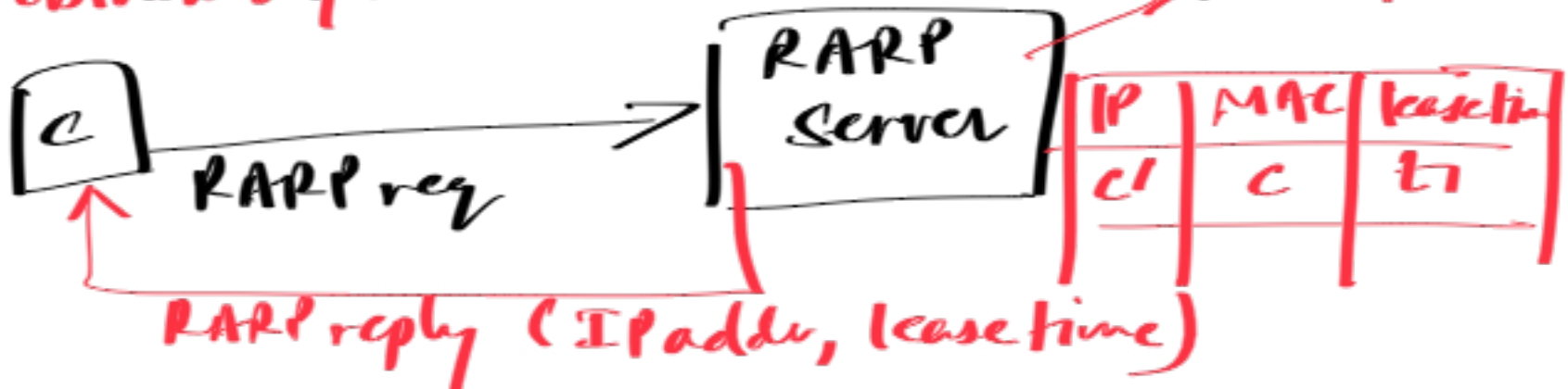
ARP req  
ARP reply (i3's MAC)

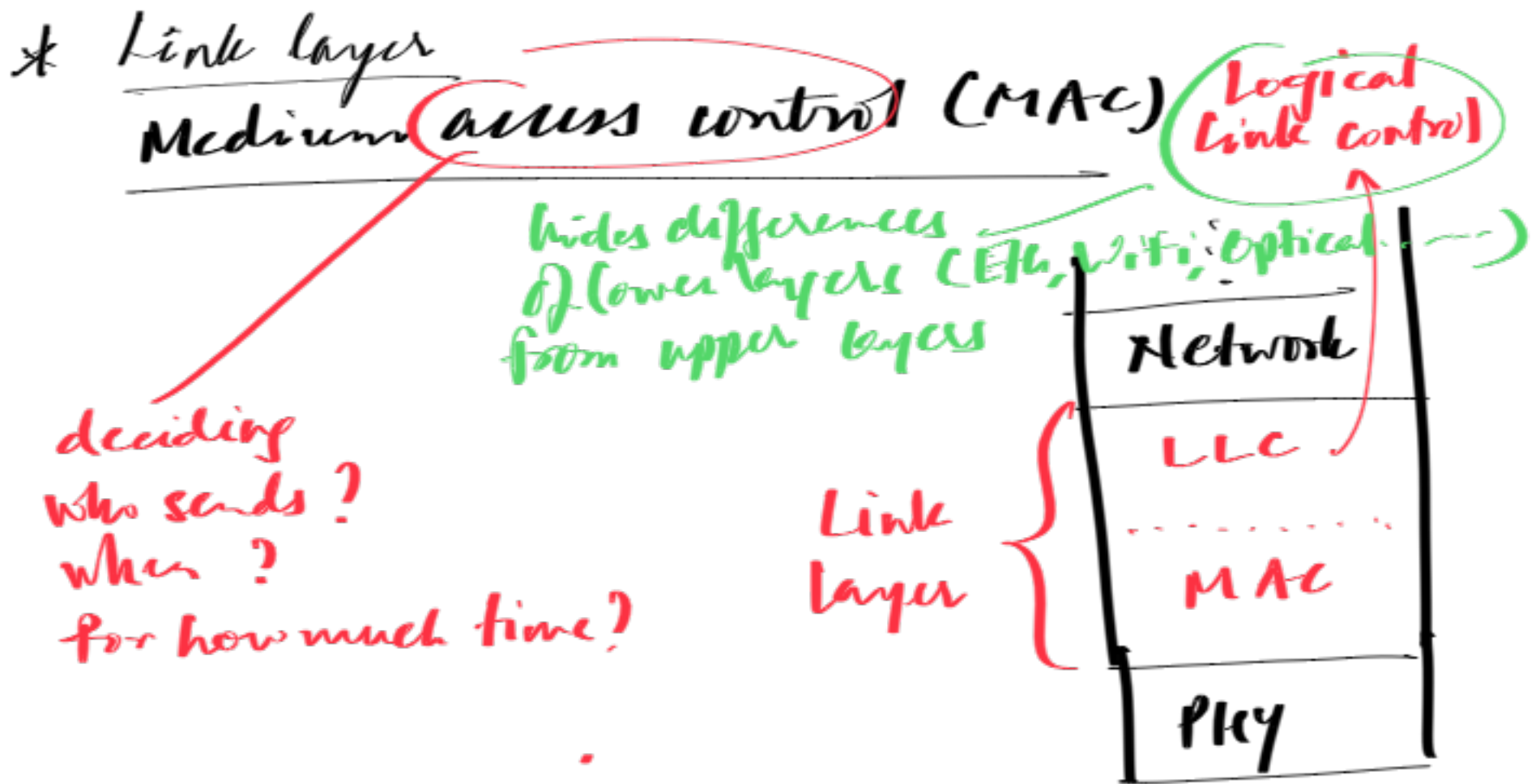
ARP req  
ARP reply (B's MAC)

other options  
- DHCP - BOOTP

\* RARP Reverse ARP maps MAC addr to corr. IP addr (self)

\* to obtain dynamic IP

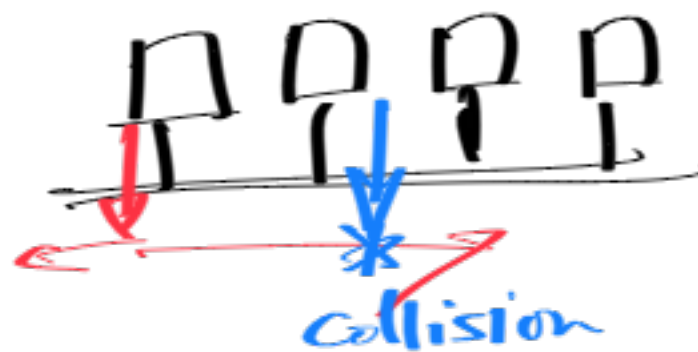




## \* MAC techniques

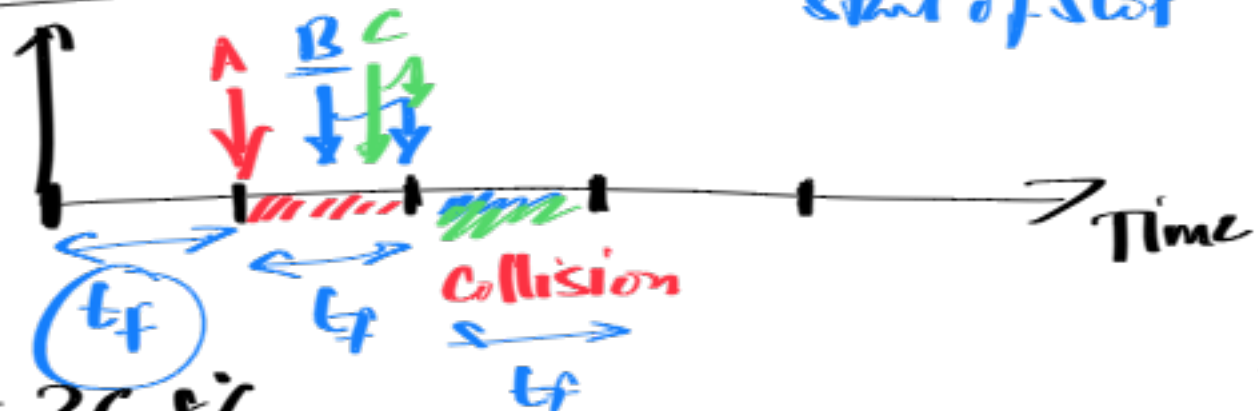
### (1) ALOHA

PURE ALOHA  
 Efficiency = 18.4%



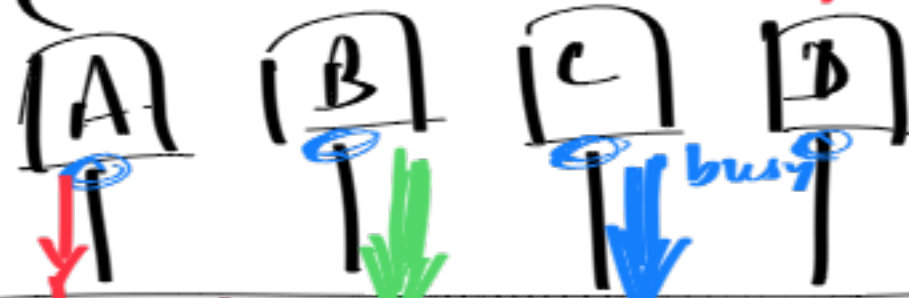
### SLOTTED ALOHA

Device can transmit only at start of slot



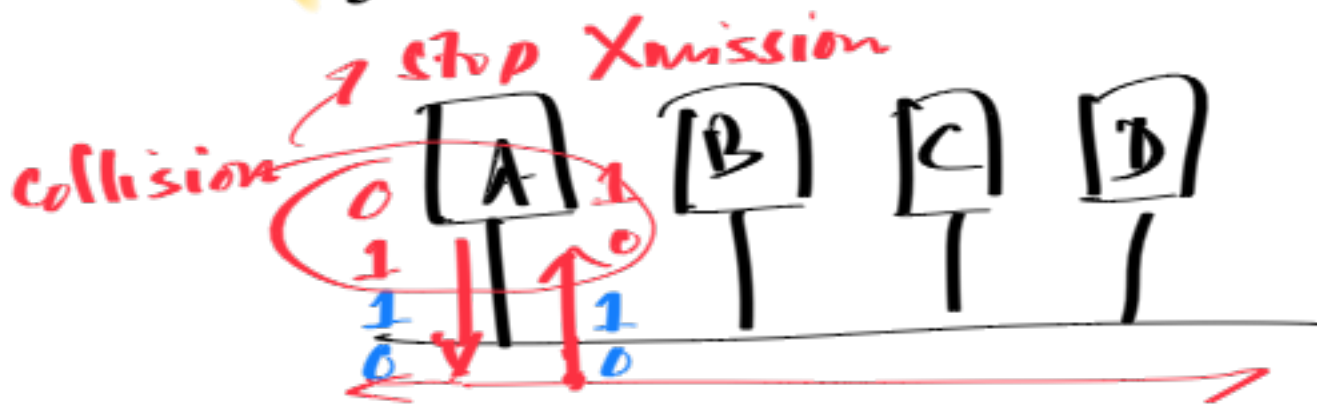
Efficiency = 36.8%

## (2) Carrier sense multiple access (CSMA) (1) persistent CSMA prob of Xmission = 1



- \* sense the channel
- idle - transmit the frame
- busy - do not send ... keep sensing the channel until idle, transmit

## (3) Carrier Sense Multiple Access with Collision Detection (CSMA/CD)



- sense the channel
- idle - transmit
  - busy -

— Collision recovery protocol  
 1. Binary exponential backoff algo.

# \* An end-to-end example

